

Question 1: Word Segmentation and POS Tagging

What you have to do

You will be given a sentence with all the spaces removed - just one long string of letters. Your program has to do two things, in order:

1. Break the string into the correct words (Segmentation).
2. Label each word with its correct grammar category — noun, verb, adjective, etc. (POS tagging).

You will do this for two languages: English and one morphologically richer language (Spanish or German — any one). The point is to see how well the same technique works on a language where word endings carry a lot of grammatical meaning (like gender and number agreement).

Example (English)

Input:

```
thequickbrownfoxjumpsoverthelazydog
```

Output:

```
[(the, DT), (quick, JJ), (brown, JJ), (fox, NN),  
(jumps, VBZ), (over, IN), (the, DT), (lazy, JJ), (dog, NN)]
```

Example (Spanish)

Input:

```
lacasarojaesgrande
```

Output:

```
[(la, DET), (casa, NOUN), (roja, ADJ), (es, AUX),  
(grande, ADJ)]
```

Notice roja (red) and casa (house) agree in gender — this is something English doesn't have, and it's exactly what makes this comparison interesting.

Step-by-Step Breakdown

Part 1 — Word Segmentation

- Train a trigram language model on your corpus. This model learns which sequences of words are common, so it can guess where one word ends and the next begins.
- Use a dynamic programming algorithm (like Viterbi) to try all reasonable ways of splitting the string and pick the most probable one.

Think of it like this: the algorithm doesn't guess randomly — it scores every possible split using the language model, and dynamic programming lets it do this efficiently instead of checking every combination one by one.

Part 2 — POS Tagging

- Train a second model that learns two things:
 - Emission probability — how likely a word is to have a certain tag (e.g., “dog” is usually a noun).
 - Transition probability — how likely a tag is to follow the previous two tags (e.g., an adjective is often followed by a noun).
- Use the same kind of dynamic programming approach to tag the segmented words.

Part 3 — Morphology-Aware Tagging

Plain POS tags (like NOUN, ADJ) don’t capture things like gender and number. Add this extra layer:

- Extend your tags to include gender/number information where relevant, e.g., NOUN-Fem-Sg, ADJ-Masc-Pl.
- This means your model now has to learn agreement patterns — e.g., a feminine noun is more likely to be followed by a feminine adjective.
- Why this matters: it lets you check something more interesting than “did I get the word right” — you can check “did the model understand grammatical agreement,” which is where English and Spanish/German really differ.

Part 4 — Baseline Comparison

Before judging your trigram+DP models, build two much simpler baselines:

- Segmentation baseline: greedy longest-match - always try to match the longest word from your vocabulary at each point.
- Tagging baseline: most-frequent-tag - just always assign the tag each word most commonly has in training data.

Report your model’s improvement over these baselines. This shows whether your model is actually worth the extra work - a key thing any real NLP evaluation should show.

Part 5 - Error Analysis with Confusion Matrix

Instead of just reporting one accuracy number, also build a confusion matrix for POS tagging - a small table showing which tags get mixed up with which (e.g., is ADJ often mistaken for NOUN?). Also check: how many tagging errors were actually caused by an earlier segmentation mistake? If a word was split wrong, its tag is almost certainly going to be wrong too - separating these two error sources gives a much clearer picture.

Small Example:

Consider the input:

`thequickbrownfox`

The correct segmentation and POS tags are:

```
the/DET quick/ADJ brown/ADJ fox/NOUN
```

Suppose the segmentation model predicts:

```
thequick | brown | fox
```

instead of:

```
the | quick | brown | fox
```

If thequick is then assigned a wrong POS tag, this is counted as an error caused by a segmentation mistake, because the incorrect word boundary caused the later tagging error.

Now suppose the segmentation is correct:

```
the | quick | brown | fox
```

but the POS tagger predicts: quick → NOUN

while the correct tag is: quick → ADJ

This is a genuine POS tagging error, because the word was segmented correctly but assigned the wrong tag.

For example, a small confusion matrix could be:

Actual Predicted	ADJ	NOUN
ADJ	60	15
NOUN	10	80

Here, the value 15 means that 15 words whose correct tag was ADJ were incorrectly predicted as NOUN. Thus, the confusion matrix shows which tags are confused, while the error-source analysis separates errors caused by segmentation from genuine tagging errors.

Corpora to Use

- English: Brown Corpus (via NLTK)
- Spanish: UD Spanish-GSD

```
git clone https://github.com/UniversalDependencies/UD_Spanish-GSD.git
```

- German: UD German-GSD

```
git clone https://github.com/UniversalDependencies/UD_German-GSD.git
```

Use the train split for training, dev split for tuning, and test split for final evaluation. For English, split Brown 80/20 yourself.

What to Submit

1. Code for segmentation model (both languages)
2. Code for POS tagging model, including the morphology-aware tags (both languages)
3. Code for the two baselines

4. Evaluation code: accuracy, confusion matrix, and error-source breakdown
5. Output examples on the sample test strings below, plus a few of your own
6. A short comparative report answering:
 - Where did English and the other language differ most in accuracy?
 - Did agreement-aware tagging actually help, or add noise?
 - How much of the tagging error came from segmentation mistakes vs. genuine tagging mistakes?
 - How much better were your models than the simple baselines?

Sample Test Strings

Spanish

- mispadrespuedenviajar → [(mis, DET), (padres, NOUN), (pueden, VERB), (viajar, VERB)]
- elcielodespejadosazul → [(el, DET), (cielo, NOUN), (despejado, ADJ), (es, AUX), (azul, ADJ)]

English

- thequickbrownfoxjumpsoverthelazydog → [(the, DT), (quick, JJ), (brown, JJ), (fox, NN), (jumps, VBZ), (over, IN), (the, DT), (lazy, JJ), (dog, NN)]

German

- autobahnmeistereiverwaltungsgebaeude → [(Autobahnmeisterei, NN), (Verwaltungsgebaeude, NN)]

Or potentially:

[(Autobahn, NN), (Meisterei, NN), (Verwaltungsgebaeude, NN)]

depending on lexicon and probabilities.

Marking Split (out of 100)

Component	Marks
Train/test split & data handling	10
Segmentation model (trigram + DP)	15
POS tagging model (emission + transition)	15
Morphology-aware tagging extension	15
Baseline models + comparison	13
Evaluation (accuracy + confusion matrix + error-source breakdown)	15
Comparative analysis report	17